

Insurance Claims Analytics

Executive Summary

Portfolio: 10,004 Policies | Analysis Period: 2023-2025 | Analytics & Insights Division

1. Business Context

The claims operations and underwriting teams requested a structured review of the current portfolio to address three operational concerns: an elevated claim rejection rate with no documented root-cause analysis, the absence of a customer risk tiering framework for renewal and underwriting decisions, and no systematic approach to identifying claims that warrant additional investigation.

This report summarises findings from a full analytical review of 10,004 policies covering five product lines. All conclusions are drawn from the available dataset. Findings involving anomaly detection and risk scoring are presented as indicators for review -- not as determinate outcomes.

2. Portfolio Snapshot

Total Policies Reviewed	10,004
Total Premium Collected	Rs 59,76,970
Total Coverage Issued	Rs 600.6M
Total Claim Exposure	Rs 1.69 Cr
Portfolio Loss Ratio	2.83x (Claim Amount / Premium)
Claims Filed	5,649 of 10,004 policies (56.5%)
Settlement Rate	33.8%
Rejection Rate	43.5%
Pending Rate	22.6%

3. Key Findings

FINDING 1 -- Rejection Rate Requires Operational Review

43.5% of all policies in the portfolio carry a Rejected claim status -- the single largest outcome category, exceeding settlements (33.8%) by 9.7 percentage points. The rejection rate is consistent across all five product lines (42.3%--46.2%) and across all age groups (42.1%--45.3%), indicating a portfolio-wide pattern rather than a product-specific or demographic issue. Without access to rejection reason codes, it is not possible to determine whether this rate reflects legitimate adjudication outcomes or processing inefficiencies.

FINDING 2 -- Premium Pricing is Undifferentiated Across Product Lines

Mean premium amounts are statistically indistinguishable across Health, Life, Auto, Home, and Travel policies (Rs 595--599 range). Coverage-to-premium ratios are similarly flat (142.6x--146.8x). A pricing structure that does not differentiate by product risk profile

exposes the insurer to adverse selection -- higher-risk policy types are effectively subsidised by lower-risk ones.

FINDING 3 -- Rs 6.8M in Pending Claims Exposure Unresolved

2,263 policies (22.6%) carry a Pending claim status, representing Rs 68,07,243 in unresolved claim exposure. Travel policies account for the largest share of pending volume by virtue of portfolio size. Pending claims that age without resolution represent both financial and customer experience risk.

FINDING 4 -- 103 Policies Flagged as High Risk for Review

A composite detection model applying six independent rules identified 103 policies (1.0%) as High Risk, with an average claim amount of Rs 5,121 -- 71% above the portfolio mean of Rs 2,994. An additional 1,714 policies (17.1%) are classified as Suspicious (1--2 flags). These records exhibit combinations of extreme claim amounts, elevated coverage utilisation, and disproportionate claim-to-premium ratios. They are candidates for review; no determination of fraud or misconduct is implied.

FINDING 5 -- No Demographic Bias Detected in Claim Outcomes

Statistical testing found no significant association between gender and claim outcome, and no significant difference in premiums paid by male vs female policyholders (Rs 597 vs Rs 598). Claim decisions appear consistently applied across gender within the available data -- a positive compliance indicator.

4. Business Risks

Risk	Severity	Driver
High rejection rate with no documented root cause	High	Finding 1
Undifferentiated premium pricing across risk profiles	Medium	Finding 2
Rs 6.8M unresolved pending claims exposure	Medium	Finding 3
103 high-anomaly policies not yet reviewed	Medium	Finding 4
Dataset lacks claim reason codes -- limits root-cause capability	Low	Data Gap

5. Recommended Actions

Immediate	Initiate manual review of the 103 High Risk policies identified by the anomaly detection model. Prioritise by total_flags count (output: anomaly_flagged.csv).
Within 30 Days	Conduct root-cause analysis on rejected claims. Obtain rejection reason codes from the claims system and re-run segmentation to identify whether rejections are concentrated in specific claim types, amounts, or filing windows.
Within 60 Days	Engage the actuarial team to review premium structures by product line. The current undifferentiated pricing does not reflect the distinct risk profiles of Health, Life, Auto, Home, and Travel portfolios.
Within 90 Days	Establish an SLA-based resolution workflow for Pending claims to reduce the Rs 6.8M unresolved exposure. Flag claims pending beyond 60 days for escalation review.
Strategic	Enrich the dataset with claim reason codes, geographic data, and customer income to improve anomaly detection precision and risk model performance in future cycles.